recognition and RFID scanning. This would make it harder for someone to steal or fake access to the vehicle because they would need to both look like the authorized user and carry the right RFID tag.

1.2 What We Already Knew RFID is used in lots of places like office ID cards, metro cards, or toll booths. It's a quick and contactless way to prove you're carrying the right thing. Face recognition is becoming more popular thanks to tools like OpenCV and Python libraries. Many phones now unlock with your face, and it's being used in smart homes and security cameras. We thought combining these two would give us a solid, low-cost way to protect car access.

We also understood that while each method has its strengths, they also have weaknesses. For example, RFID cards can be stolen or copied, and face recognition can sometimes be fooled by a photo. By requiring both, we greatly reduce the chances of someone getting in who shouldn't.

1.3 What Problem We're Solving The main problem is that cars don't have strong enough security. A single method, like a key, can be easily stolen. Even face ID systems can sometimes be tricked using photos. Our project solves this by adding a second step, scanning an RFID card. Only when both steps are passed will the "car" which is simulated by a green LED turns on. This layered approach makes it harder for attackers to break in.

1.4 How We Did It We broke the project into two big parts:

- The first part is face recognition, which we built using Python and the OpenCV library. It runs on a laptop and checks if your face is the one it knows. If it finds a match, it sends a signal to the second part of the system.
- The second part is RFID scanning using an Arduino and the MFRC522 RFID module. If your face is verified, the Arduino then asks you to scan your RFID card.

If both steps are successful, a green LED turns on to show that you're allowed to start the car. If either step fails, access is denied. We chose this design to mimic how some real-world systems work: one device handles biometric data, while another takes care of card-based access.

1.5 What We Expected to Happen We were hoping that:

- The face recognition system would detect someone's face from about 3 to 4 feet away, even in normal lighting conditions.
- The RFID tag would scan and respond quickly, ideally in under 5 milliseconds.

Smart Vehicle Ignition system with RFID & Face recognition

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Abstract

Our project is about building a smart car ignition system that uses both face recognition and an RFID card to allow the car to start. The reason for doing this is that most cars today still rely on just a key or a key fob, and those can be stolen or copied. To make car access more secure, we created a two-step process. First, the car checks if your face matches the one saved in the system. Then, it asks you to scan an RFID card. Only if both your face and the RFID card match, the car gives the green light to start, in our case, a green LED lights up.

We used Python and OpenCV for the face recognition part. The RFID card reader was set up using an Arduino Uno with an MFRC522 module. The two systems are connected in sequence, face recognition happens first, and only if it succeeds, the RFID part becomes active. If everything checks out, an LED lights up to show that access is granted.

We tested both parts individually and got great results. The face recognition worked well from about 3 to 4 feet away, and the RFID card was scanned in just a few milliseconds. We also learned a lot about security, coding, and working with hardware. In the future, we'd like to add more security features like liveness detection to stop people from fooling the system with photos, and maybe even let it control an actual car ignition instead of just lighting up an LED.

1. Introduction

1.1 Why We Wanted to Do This Project Cars today are still pretty easy to steal if someone gets your key or key fob. We wanted to build a smarter and safer way to start a car, something that checks who you are using your face and also makes sure you have the right card. This idea of using two different things to confirm your identity is called "multi-factor authentication," and it's used in apps, banking, and more. So, we thought, why not bring that to cars too?

Our goal was to build something simple but powerful using low-cost components. Instead of trying to reinvent everything, we decided to combine two technologies that already exist: face

- If someone who wasn't authorized tried to use it, the system would clearly say no and not activate the LED.
- We'd get visual feedback, green light for success, red light for failure, to make testing easier and clearer.

We also hoped the system would be easy to test and modify. That's why we made sure each part could run on its own before trying to connect them. That way, if something didn't work, we'd know exactly where the problem was.

1.6 How This Connects to Our Class In class, we learned about how systems can be made more secure through multi-layered protection. We applied those ideas directly here: first using face recognition , "something you are", and then using RFID, "something you have". This is a real-world example of what we discussed about system threats, access control, and making smart decisions about security design.

We also followed good security design practices from the course, like thinking about the weakest link, how attackers might exploit flaws, and how to give clear feedback to users. This project gave us a chance to take those ideas and apply them using real hardware and software.

1.7 What's in the Rest of This Report This report walks you through everything we did. First, we'll go over other similar projects and papers (related work). Then we'll explain what tools and code we used. After that, we'll describe who did what and when. We'll show our results, reflect on what went well and what didn't, and end with what we'd do next to make the project even better.

Related Work and Additional paper :

1.M. Pathak, K. N. Mishra, S. P. Singh, and A. Mishra, “An automated smart centralised vehicle security system for controlling the vehicle thefts/hacking using IoT and facial recognition,” in *Proc. 2023 Int. Conf. Comput. Intell. Knowl. Economy (ICCIKE)*, Dubai, United Arab Emirates, 2023**,**

Overview**:** Raspberry Pi-Based Intelligent Car Anti-Theft System through Face Recognition Using GSM and GPS

This project presents a smart, low-cost, and reliable vehicle anti-theft system that uses facial recognition, GSM, and GPS technologies, with a Raspberry Pi as the central controller. The system is designed to enhance vehicle security by ensuring that only authorised individuals can start the car, while providing real-time alerts and tracking in the event of unauthorized access.

The system begins by using a camera connected to the Raspberry Pi to capture the face of the person attempting to start the vehicle. Facial recognition is performed using computer vision libraries such as OpenCV or face_recognition, which extract facial features and compare them against a database of pre-authorised users. If the face matches, a relay module is activated, enabling the vehicle’s ignition system. If the face does not match any authorised profiles, the system takes immediate action by blocking ignition and sending a real-time SMS alert to the vehicle owner via a GSM module (such as SIM800L).

Along with the alert, the system uses a GPS module (e.g., NEO-6M) to fetch the current location of the vehicle and includes a Google Maps link in the message, allowing the owner to track the vehicle instantly. This dual-layered security—biometric verification and location-based alerting—offers a significant improvement over traditional key-based or RFID-based systems, which can be easily lost, stolen, or cloned.

Because the system operates without requiring an internet connection, it is especially suited for deployment in remote or low-connectivity areas. It combines hardware components (Raspberry Pi, camera, GSM, GPS, relay) with software algorithms (face detection, encoding, and matching) to create an intelligent, autonomous security solution. The proposed system not only prevents unauthorized access but also provides live monitoring and recovery support, making it an ideal solution for both personal and commercial vehicle security.

2. Deokar and S. More, "Face Detection Based Vehicle Ignition System," in *Proceedings of the 5th International Conference on Communication and Information Processing (ICCIP)*, 2023, pp

This paper proposes a smart vehicle security system that uses face detection technology to control vehicle ignition, aiming to reduce vehicle theft and unauthorised access. The system relies on computer vision to verify the identity of the driver before allowing the ignition process. When a user attempts to start the vehicle, a camera captures a real-time image of their face. This image is then processed using facial detection algorithms to determine if the user is authorised.

The primary hardware used in the system includes a Raspberry Pi or similar processing unit, a camera module, and a relay circuit to control the ignition. The system employs OpenCV for face detection, which identifies key facial features and compares them with pre-registered data. If the face is successfully detected and matched, the system activates the ignition relay, allowing the car to start. If the face is not recognised, the ignition remains disabled, thus preventing unauthorised use.

This solution offers a contactless, biometric-based authentication method that enhances vehicle security without relying on traditional keys or RFID cards, which can be lost or duplicated. The system is designed to be low-cost, scalable, and suitable for both personal and commercial vehicle applications. It also provides a foundation for integrating additional features such as GSM alerts, GPS tracking, or multi-factor authentication.

Overall, the paper demonstrates a practical and efficient approach to secure vehicle access using facial recognition, highlighting the growing role of AI and computer vision in automotive security systems.

3.. H. Lemelson and L. J. Hoffman, *Vehicle security system using facial recognition*, U.S. Patent 7,116,803, issued Oct. 3, 2006. This patent is a continuation of U.S. Patent 6,831,993, issued Dec. 14, 2004, which is a continuation of U.S. Patent 6,400,835, issued Jun. 4, 2002, based on application Ser. No. 08/647,674, filed May 15, 1996.

This system introduces a biometric-based approach to vehicle security by using facial recognition technology to control access and operation of the vehicle. Instead of relying solely on physical keys or traditional security systems, the proposed solution integrates a facial-recognition scanner, such as a television or infrared camera, positioned to capture the face of the person in the driver's seat. The captured image is processed by a facial-recognition computer, which determines whether the individual is authorised. Based on the result, control signals are generated to either enable or disable critical vehicle functions, such as the ignition or starter motor.

One notable design described in the system involves placing the camera within the third taillight assembly, typically located inside the rear window and below the roofline. This configuration allows the camera to face forward toward the rear-view mirror, providing a clear view of the driver's face. The system can work independently or in conjunction with other security measures and may also include features that allow automatic vehicle starting once the user is authenticated.

Overall, this approach provides a secure, contactless, and intelligent solution for enhancing vehicle security through real-time facial identification.

4.C. Kumar G. S. and Y. R., “Facial-recognition vehicle security system,” *Int. J. Sci. Res. Eng. Trends*, vol. 3, no. 10, pp. 1–7, Oct. 2021

This paper presents a biometric-based vehicle ignition system using face recognition technology to enhance car security and reduce the risk of unauthorised access. The proposed system eliminates the need for traditional keys or RFID cards by employing real-time facial recognition to authenticate the driver before allowing the vehicle to start. Developed using image processing techniques, the system captures the facial image of the user via a camera installed in the car. It compares the captured image with a pre-stored database of authorised users using facial recognition algorithms. If a match is found, the system activates the ignition circuit; otherwise, access is denied.

The system is implemented using Python with OpenCV for face detection and recognition, and integrated with hardware components such as a Raspberry Pi or similar microcontroller, a camera module, and a relay circuit to control the vehicle's ignition system. The approach focuses on real-time authentication, contactless operation, and ease of use, making it a practical and cost-effective solution for personal and commercial vehicle security. Overall, the project aims to provide a secure, intelligent, and user-friendly ignition system, replacing conventional methods with biometric verification to improve vehicle safety and user experience.

Methodology and Tools:

Tools needed:

- Arduino Uno
- Jumper wires
- Mifare RC522 RFID 13.56MHz kit
- Computer that has webcam built into it
- Breadboard
- Green and Red LEDs
- 1K ohms resistor

Software needed:

- Python
- Arduino IDE
- Face_recognition package
- Cv2 package

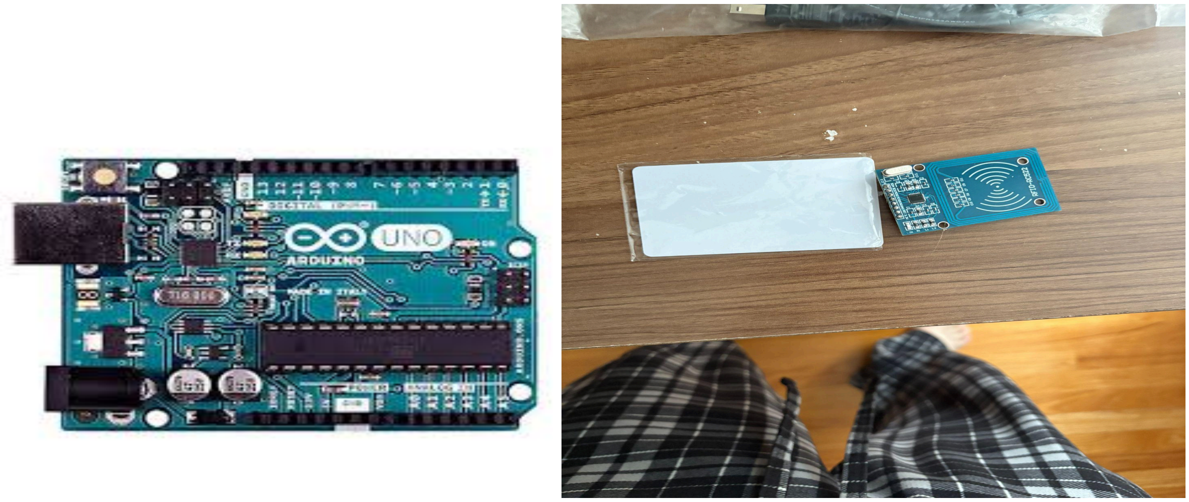

Figure 1: Arduino Uno(left) and the RFID scanner with it's card(right)

Steps to build:

1. Get the Arduino Uno, breadboard, RFID scanner, and jump wires from somewhere
2. Put the RFID scanner on the breadboard and use the jumper wires to connect it to the Arduino Uno. SCK connects to pin 13 of Arduino Uno, MISO connected to pin 12, MOSI connected to pin 11, SDA connected to pin 10, and RST connected to pin 9 of the Arduino Uno. Make sure that the RFID scanner is connected to a GND and a 3.3 V
3. Put the Green and the Red Leds on the breadboard where their anode(the longer leg of the LED is connected to pins 7 and 6, respectively of the Arduino Uno and connect their cathode to the ground, which is going through a 1k resistor with a jumper wire.
4. Install face_recognition and the cv2 to be able to do the face recognition in Python with the computer's camera, and create a folder in Python where it has the user's faces that are authorised, which the program can access.

5. Create a program where it uses those in step 4 where once it finds an authorized user with the camera that was in the folder, it then allows the user to proceed to use the RFID scanner, where it signals a signal to proceed to read the RFID card/tag.
6. Connect the the Arduino to your computer and open up the Arduino IDE to check what is the UID that card/tag has it on it so you would add it to the authorized UID list later on by using the MFRC522 library
7. Create a program in the IDE where it waits for the signal from the Python that is able to recognize a face then it starts to read the RFID tag where it would light up green or red LED depending on if the RFID reads a authorized UID from the card/tag where once done it would where it is first uploaded to the Arduino.
8. If it is able read the authorized code, the arduino would send a signal to the Python to show that it was verified and that it could exit out of the program thus shutting the camera down and showing that it would allow the authorized user to go through the 2 securites.

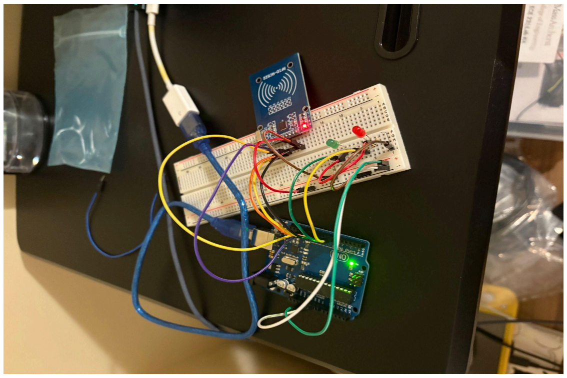

Figure 2: Finish build

Partitioning of Work and Schedule :

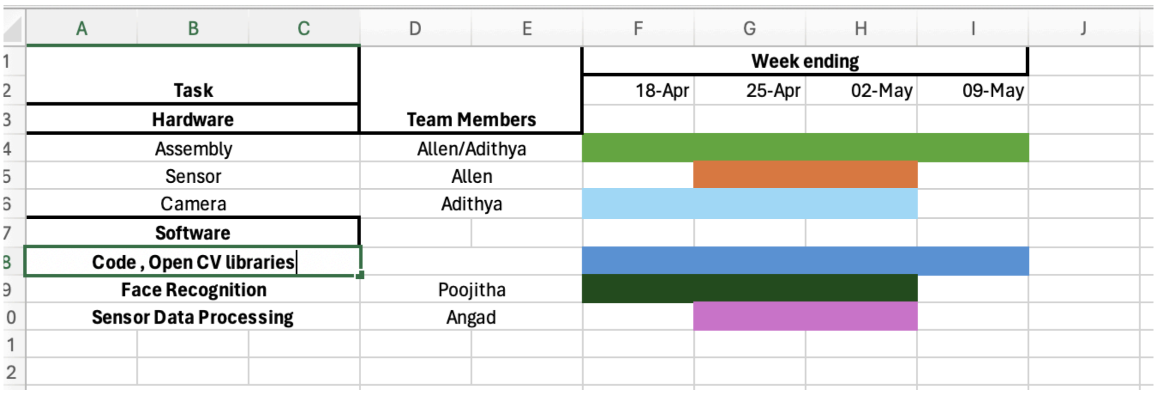

Figure 3:The Tasks were divided according to the Gantt chart

Allen : I decided to work on the hardware along with Adithya where we worked together putting the hardware together to be able to show the demo. I was responsible for procuring the hardware needed to create the project. I also helped test to see if the Ardunio is able to function properly with the face recognition and the RFID scanner working together to create the 2-factor authentication when the software part of our group shared the code with all of us to see if it is able to work.

Adithya: As part of the project, along with Allen, I was responsible for the design and methodology of the project. The assembly of the circuit along with the block diagrams it required and the ideologies behind the thought process was architected. I also helped set up the camera using the webcam on our laptop with OpenCV for the purpose of the facial recognition section of our project.

Poojitha : As part of the project, I was responsible for developing and implementing the facial recognition module that controls vehicle ignition based on user identity. A USB webcam connected to an Arduino and was used to continuously capture real-time video frames of the person sitting in the driver's seat. Using Python and the face_recognition library, which utilises the library for facial landmark detection, I built a system that first encodes a reference image of the authorised user.

During operation, the system detects faces in the live feed, encodes them, and compares the result to the stored encoding using Euclidean distance. If the match is successful within the predefined threshold, the system initiates communication with the Arduino Uno via USB using the pyserial library. A control signal ('1' for authorised, '0' for unauthorised) is sent.from the code to the Arduino.n this project, I used the face_recognition, OpenCV, serial, time, and os libraries to implement a secure facial recognition-based vehicle ignition system integrated with Arduino.

The code first establishes a serial connection with the Arduino and loads known face images from a local folder, encoding them for comparison. It then starts the webcam using OpenCV and continuously captures video frames. For each detected face, it checks if it matches any of the pre-stored encodings. If a match is found, the system sends a "face_ok" signal to the Arduino over USB serial. The Arduino is then expected to verify an RFID check and respond with "rfid_ok". If both verifications succeed within a 10-second timeout, the system exits and allows ignition. Otherwise, it continues scanning. The program also includes basic error handling

Figure 2.0 Code for Facial Recognition.

```python
import cv2
import face_recognition
import serial
import time
import os

# --- Setup Serial Connection ---
try:
    arduino = serial.Serial('/dev/tty.usbmodem101', 9600)
    time.sleep(2)  # Allow Arduino to reset
    print("✅ Serial port opened successfully")
except serial.SerialException as e:
    print(f"❌ Could not open serial port: {e}")
    exit()

# --- Load and Encode Multiple Known Faces ---
known_face_encodings = []
known_face_names = []

face_folder = "known_faces"

for filename in os.listdir(face_folder):
    if filename.endswith((".jpg", ".png")):
        name = os.path.splitext(filename)[0]
        path = os.path.join(face_folder, filename)
        image = face_recognition.load_image_file(path)
        encoding = face_recognition.face_encodings(image)
```

Figure 4 Code for Facial Recognition

```
✅ Recognized Allen – LED ON
✅ Recognized Allen – LED ON
✅ Recognized Allen – LED ON
✅ Recognized Allen – LED ON
✅ Recognized Adithya – LED ON
✅ Recognized Adithya – LED ON
✅ Recognized Allen – LED ON
✅ Recognized Allen – LED ON
✅ Recognized Allen – LED ON
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✅ Recognized Allen – LED ON
✅ Recognized Allen – LED ON
^Z
```

Figure 5 Code detected Allen's face

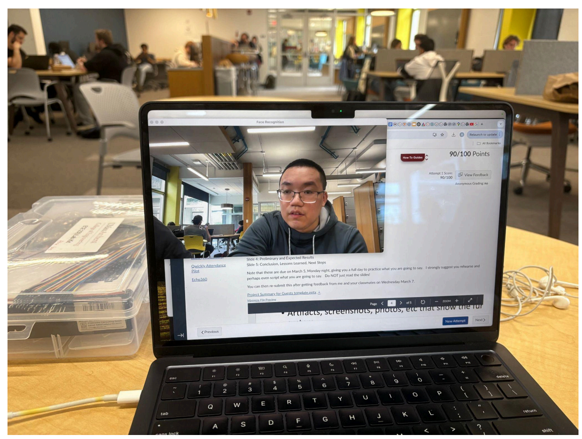

Figure 6 Face is captured

Angad Deep Singh : As part of this project, Angad Deep Singh was responsible for the RFID authentication system, which formed the second step in our dual-authentication smart ignition setup. From the start, once the team decided on a two-stage system, facial recognition followed by RFID, I focused entirely on designing, building, and testing the RFID side using Arduino Uno and the MFRC522 RFID reader module.

I began by understanding how the RFID hardware works and how it communicates with the Arduino through SPI. I spent time reading documentation, watching tutorials, and testing example code. I tested how well the RFID reader picked up tags from different distances and angles.

Once the basic reading functionality was working, I developed a custom Arduino sketch that compares each scanned UID with a list of authorized tags. I added visual feedback using two LEDs: green for access granted, red for access denied. This helped simulate whether the ignition would be allowed to proceed.

Toward the end of the project, I modified the code so that the RFID reader would only activate after the facial recognition module had successfully identified a user. This helped us build the two-step process we were aiming for. Through this work, I strengthened my skills in embedded systems, Arduino programming, and thinking through real-world security challenges.

Experiments and Results

In the project, we created a mock smart vehicle ignition system with RFID scanner and face recognition where we connected the Arduino Uno where we created 2 programs for it. One of the programs does face recognition which is runned with Python where it sends a signal to the Arduino once it finds a face it recognizes from the pictures that are in the folder that the program grabs from that are authorized with the computer's camera. The arduino contains the other program that was done in Arduino IDE where it has a program where it would start to allow the user to start to scan their RFID card/tag to the scanner where it checks to see the UID that it is authorized once it receives that the signal from the python program that the face recognition was successful. Once both the RFID and the face recognition got the correct inputs, the arduino would light up the green led to indicate that the two security were successful and exit out of the program. However, if the face recognition was able to go through but not the RFID, then the Red led would light up to indicate that and still wait for the card that contains the right UID. If the camera doesn't recognize the face that isn't in the folder that contains the folder of the authorized user, then the terminal would say that it can see the face but doesn't recognize the face as their face isn't in the folder that allows the user to be recognized by the camera.

The camera's recognition distance was tested to see how far away it can see a person's face. The drawing below shows how far the camera could still read a person's face and recognize them from the authorized user's face. It shows that it is able to see a person's face up to around 20 feet away from the camera before it is unable to see the person's face once it goes beyond around 20 feet away. It was easily tested by simply having a person keep taking some steps away from the camera and wait and see if it can still recognize the person from that distance until it is unable to see their faces.

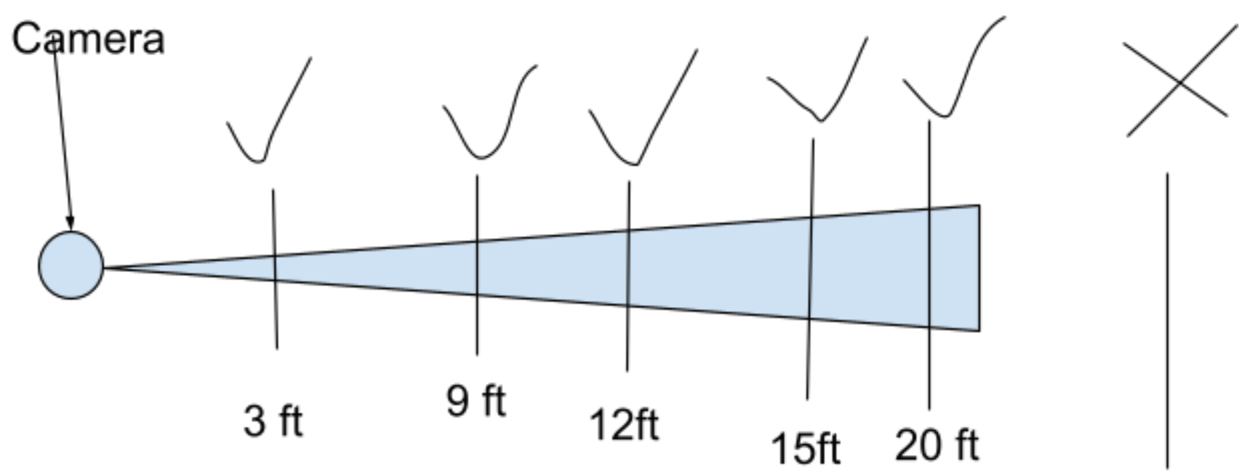

Figure 7: Camera's viewing distance

Another thing that was tested with the system to see how fast the RFID scanner is able to read the card to get its UID. So to test it we had the arduino system to itself to see how fast it could be able to get the UID from the card each time it is scanned.

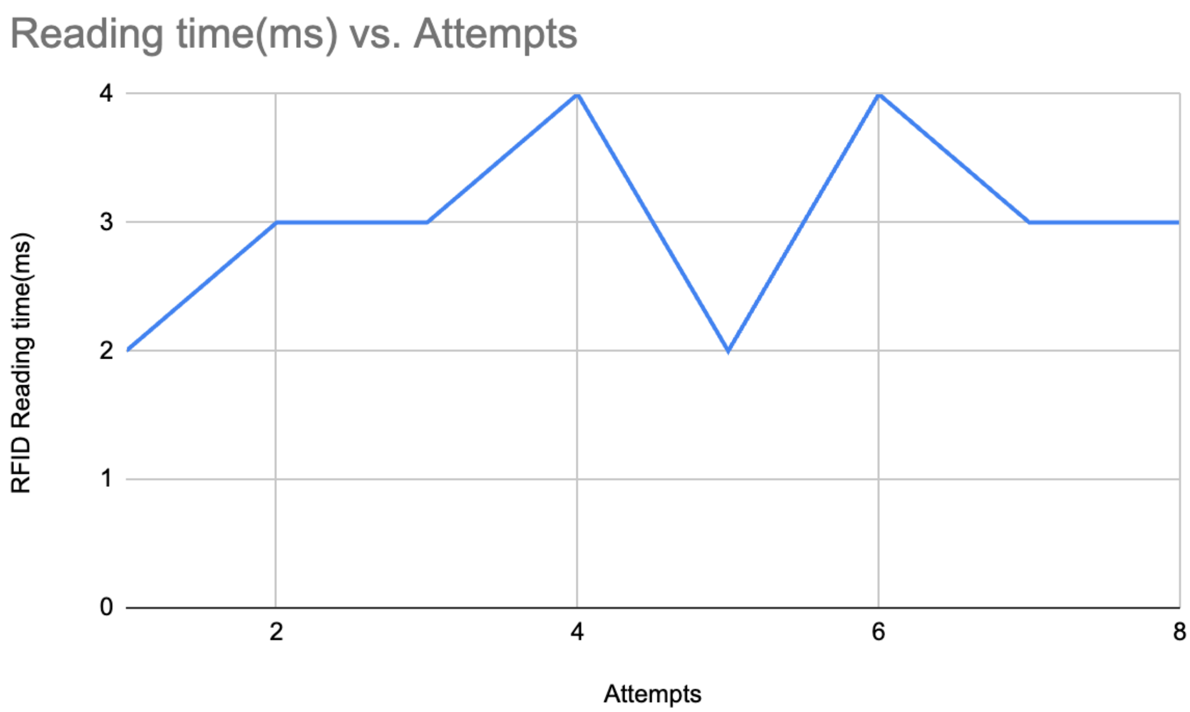

Figure 8: RFID reading time per attempts

The graph above shows how fast the RFID scanner can read the UID from the card. From the graph, the lowest time it could read the card is about 2 ms and the longest time that the RFID scanner could take to scan the card is about 4 ms. However, from testing it, the scanner is able to read the card at most in about 3 ms because most of the attempts done, the system reads that it can do it at 3 ms in the serial monitor.

Camera was tested to see if it is able to be fooled by a photo of the user. And so a photo of one of our members was used to test to see whether or not the camera would be fooled by it and allow them to bypass the face recognition system.

Unfortunately as shown by the graph below, the program is easily fooled by the picture of the authorized user which is stated earlier as an attack vector as the live face and photo of them have the same exact success results from the facial recognition. They both have the same success rate with being recognized at 100% rate.

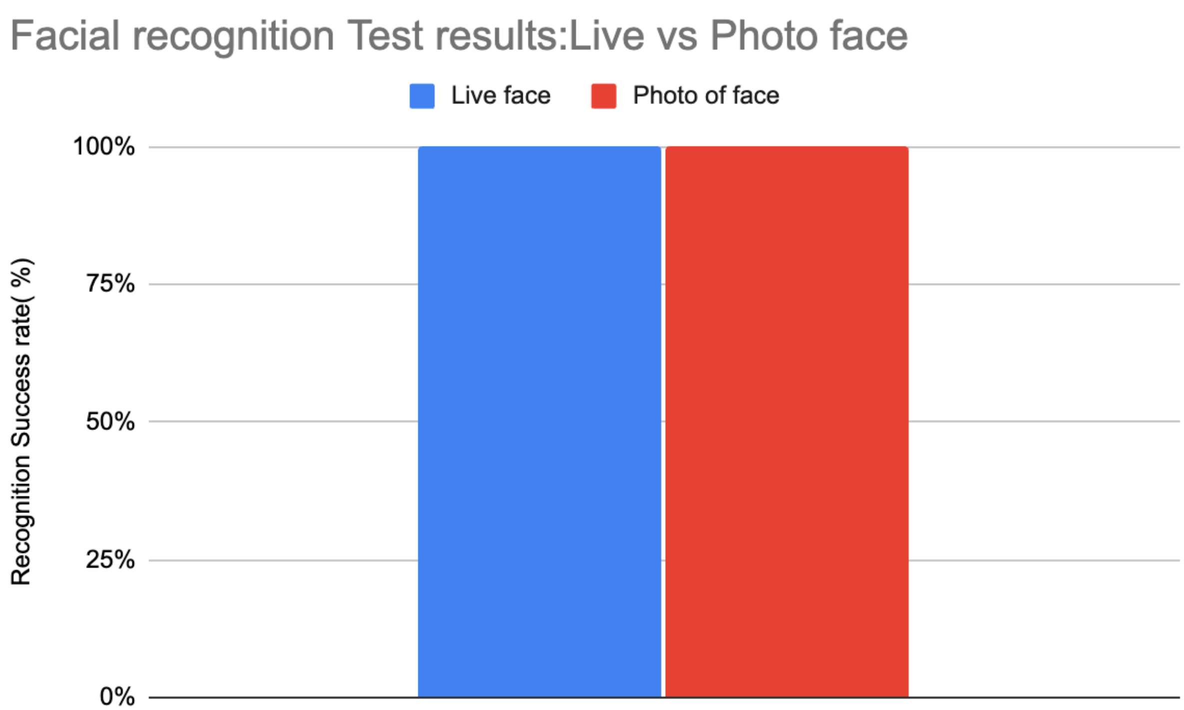

Figure 9: Live vs Photo face recognition

Lastly, we decided to test the camera again by checking to see if it is able to recognize the user's face if they are wearing hats,glasses,face mask, scarfs or anything that could obstruct their faces from the camera. So we decided to test it by wearing things near our faces to obstruct them like putting an arm covering half our face like a scarf to see if the camera and still detect our faces our can't.

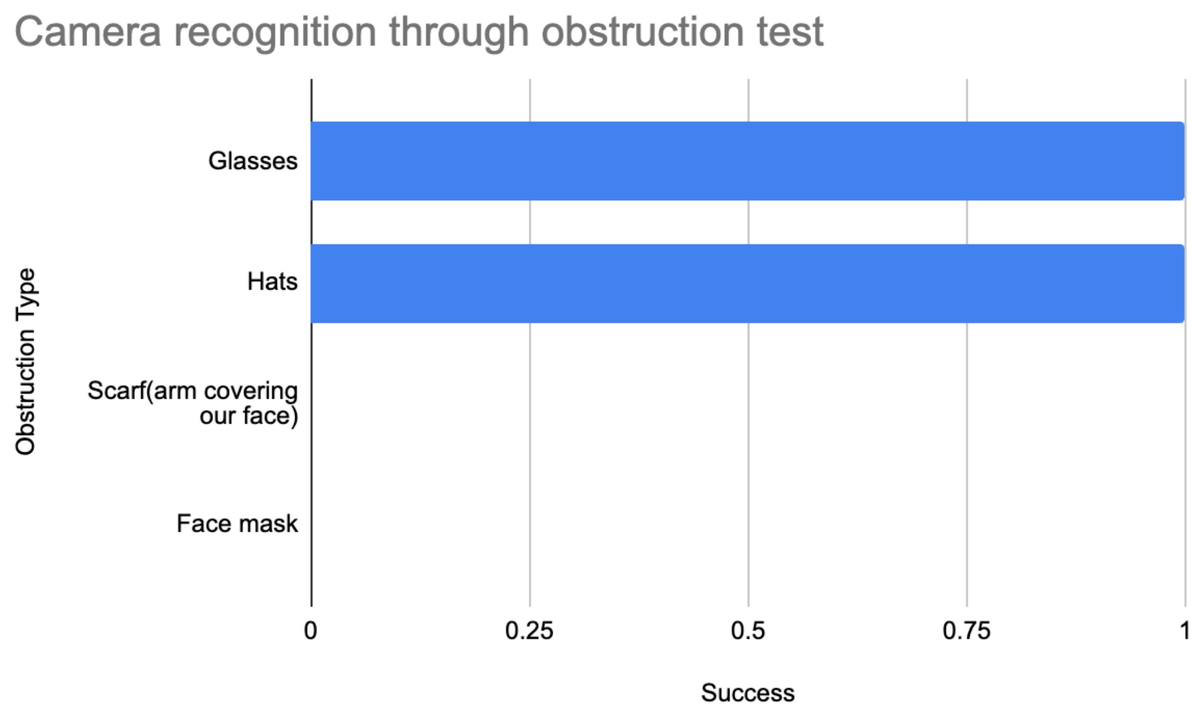

Figure 10: Recognition through Obstruction

The graph above shows whether or not the camera is still able to recognize the user's face with these obstructions. The graph shows that the camera is able to easily recognize the user when they are wearing a hat,glasses or both of them. However, the test helped discover that the camera isn't able to recognize a user's face if they are wearing a scarf or a face mask that covers half the bottom of their faces which makes it unable to get to the RFID scanner if their face are obstructed.

Link to the demo

https://youtu.be/OIoVZKDeuDg

Analysis and reflections

From a technical standpoint, we gained hands-on experience with both hardware integration and software logic design. Working with the RC522 RFID module, we understood how identity verification using RFID tags works at a low level, including UID reading, SPI communication, and basic encryption concepts. On the facial recognition front, using OpenCV and face_recognition API taught us how computer vision systems can be practically deployed for biometric authentication. We learned the nuances of working with live image input, training datasets, and matching embedding vectors for facial verification. We also learned how to simulate vehicle ignition using visual feedback using LEDs.

Realizing the value of multi-layered security was one of the most important lessons learned. The studies we looked at made clear that using facial recognition or RFID alone is insufficiently reliable for contemporary security applications. In order to successfully deploy two-factor authentication (2FA) for physical access control, we recognized the benefits of combining both RFID for user identity confirmation and facial recognition for biometric confirmation

Additionally, we exercised actual project management approaches, such as creating block diagrams, Gantt charts, and effectively dividing workstreams. We also learned the challenges of hardware-software integration.

What Could We Have Done with One More Week?

With just one more week, we could have moved beyond simulation and demo mode toward building a fully enclosed prototype. While we managed to wire components and test them on breadboards, creating a compact casing with a well-organized internal circuit would have made the project look more professional and real-world-ready.

Additionally, we could have refined our testing framework. With extra time, we would have expanded the test suite to include:

- More faces under different lighting conditions
- False rejection and false acceptance testing
- Environmental robustness checks (e.g., camera delay under low power)
- Documentation and UI: More time could also be invested in developing a cleaner user interface and producing a technical manual

What More Could We Have Done with One More Month?

- **Mobile Integration**: Adding SMS alert and ignition control features would allow remote disabling or enabling of the vehicle via a phone.
- **Liveness Detection**: Face recognition systems can be spoofed using photos. With additional time, we could have implemented basic liveness detection using blinking or motion cues to ensure that only real users pass verification.
- **Encryption and Data Privacy**: A longer timeline would have allowed us to integrate encryption protocols for storing face data and tag IDs securely

What Additional Skills or Expertise Would Have Helped?

- Security Protocol Design: Understanding and implementing secure RFID protocols, such as mutual authentication, could have prevented tag cloning. Similarly, anti-replay attacks and secure communication between the modules were areas we could have benefited from with more cybersecurity background.

- Embedded Systems Experience: More embedded C/C++ or Arduino IDE knowledge could have helped us optimize hardware usage.

- RFID Tag Security Analysis: Knowing how to prevent RFID spoofing or cloning would enhance system robustness.

- Biometric Data Privacy: Familiarity with general biometric data protection principles could guide secure data storage, especially if facial data is stored or transmitted.

7. Conclusion and Future Work

7.1 Conclusion In this project, we built a prototype for a Smart Vehicle Ignition System that uses both face recognition and an RFID tag to allow someone to start a car. The goal was to make sure only the right person, someone whose face is known and who has a specific RFID card, can start the engine. This way, even if someone steals the car keys or gets hold of the RFID card, they still can't start the vehicle unless their face is also recognized by the system.

We used two separate systems: a face recognition system running on a laptop or Raspberry Pi, and an RFID scanner connected to an Arduino Uno. When a face is recognized successfully, the Arduino system is told to activate and wait for a valid RFID scan. If the tag is recognized too, the system lights up a green LED to show the car is now ready to start. If either the face or RFID scan fails, a red LED lights up, and access is denied.

Working on this project taught us a lot about how to combine software and hardware. We learned how RFID works, how to scan and compare UID tags, how to use Python libraries for facial recognition, and how to program and debug Arduino systems. Most importantly, we learned how to think like security engineers, how to identify weaknesses (like spoofing with a photo) and try to make the system more secure by using two different ways to confirm identity.

Even though both parts of the system worked independently, we were still in the process of fully combining the two systems into one smooth user experience by the time we presented our results. But overall, the system showed that with simple tools and a little creativity, we can make vehicle access much more secure than traditional key systems.

7.2 Future Work Looking ahead, there are many improvements we would like to make. The first step would be to complete the integration of both systems so that the face recognition and RFID check happen seamlessly within one user-friendly program. This would allow the whole system to work automatically, with no manual steps needed between face scanning and card scanning.

Another big upgrade would be adding liveness detection to the face recognition system. Right now, someone could try to trick the system using a printed photo or video. Liveness detection uses things like blinking or head movement to make sure a real person is standing in front of the camera.

We would also like to expand the system to support more than one user. That means allowing multiple faces and RFID tags to be registered and managed in the system. Additionally, we could improve the security of the RFID component by encrypting the UID values instead of storing them in plain text.

Eventually, we'd like to move from using just an LED to simulate ignition, and instead connect the Arduino to a relay switch that can control the real ignition line in a vehicle (or a model). With more time, we could test the system in different real-world conditions, like in a moving car, in low light, or with more people in the camera frame, to see how robust it really is.

In the long term, the system could be applied to other use cases too, like smart home locks, office entry systems, or even industrial equipment. It's a great example of how basic security ideas from class, like multi-factor authentication and layered defence, can be turned into something that's both practical and impactful.